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VLM-AR3L

Dual VLM Reward Signals—Absolute and Relative—Unlock Open-World Reinforcement Learning Without Reward Engineering

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Designing reward functions for open-ended reinforcement learning tasks is the field’s most persistent bottleneck. Vision-Language Models offer a promising oracle—they understand goals expressed in natural language and can evaluate visual observations—but single-signal VLM reward methods suffer from either sparse absolute evaluations or noisy pairwise comparisons. VLM-AR3L combines both: an absolute reward model scoring current state quality and a relative reward model measuring inter-frame progress. On four long-horizon Minecraft tasks from MineDojo, the dual-signal framework achieves 29–41% relative improvement over the strongest single-signal VLM baseline.

AT A GLANCE

Problem: Absolute VLM rewards are sparse in long-horizon tasks; relative VLM rewards lack goal-distance information. Neither alone trains effective agents.

Why it matters: Open-ended embodied RL requires dense, well-shaped reward signals without human-designed reward functions.

Method: Separate absolute (scalar state evaluation) and relative (pairwise progress) reward MLPs trained on offline VLM preference labels; combined as weighted sum for PPO policy training.

Result: 29–41% relative improvement over best single-signal VLM baseline on all four MineDojo Minecraft task categories.

Evidence: Ablation confirms both reward branches contribute; removing either degrades success rate by 20–30%.

The Big Picture

Reinforcement learning in open-ended, visually complex environments requires reward signals that are dense enough to provide learning signal throughout each episode and semantically grounded enough to evaluate progress toward linguistically-expressed goals. Human reward engineering meets both requirements but is expensive and

brittle—reward functions designed for one task rarely transfer.

Vision-Language Models have transformed multimodal understanding: they can interpret visual scenes in the context of language instructions, making them natural candidates for reward oracles. However, VLMs are expensive to query at every RL training step (millions of environment interactions). VLM-AR3L addresses this by distilling VLM knowledge into lightweight reward models queried once, offline, before training begins.

Why Existing Approaches Fall Short

EUREKA and similar methods generate only *absolute* scalar rewards: the VLM evaluates each frame independently and assigns a score. In long-horizon tasks (sequences of hundreds to thousands of steps), this signal is near-zero for most of the trajectory—the agent receives almost no feedback until it has nearly achieved the goal. Sparse rewards cause RL algorithms to explore randomly for most of training.

Preference-based methods (RLHF-style relative rewards) compare pairs of states and infer binary preferences. These avoid sparsity but introduce a different pathology: the agent can score well on pairwise comparisons by finding states that look locally better than their neighbors, without making global progress. This leads to oscillatory behavior around local optima.

Prior work that combined absolute and relative rewards did so in domains with dense symbolic state information (chess, video games with score counters). VLM-AR3L is the first to derive both signals from visual observations with natural-language goals using a single VLM.

Core Mechanism

The framework decomposes the reward signal into two complementary streams:

Absolute Reward Model (ARM): A learned scorer $f_{\phi}^{abs} : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps a current visual observation x_t and language goal g to a scalar task-achievement estimate. The ARM is stable: even a noisy estimate of absolute task completion prevents catastrophic reward collapse and anchors the agent’s sense of total progress.

Relative Reward Model (RRM): A learned comparator $f_{\psi}^{rel} : \mathcal{O} \times \mathcal{O} \times \mathcal{G} \rightarrow [-1, 1]$ compares consecutive frames (x_t, x_{t+1}) in the context of goal g to infer local progress or regression. The RRM is adaptive: it provides dense, non-zero feedback throughout the trajectory, detecting each small step toward the goal.

Technical Formulation

The combined reward at step t :

$$r_t = \alpha \cdot \sigma(f_\phi^{abs}(\text{enc}(x_t), \text{enc}(g))) + (1 - \alpha) \cdot \tanh(f_\psi^{rel}(\text{enc}(x_t), \text{enc}(x_{t+1}), \text{enc}(g)))$$

where $\text{enc}(\cdot)$ denotes a frozen CLIP encoder. The ARM is trained by minimizing squared error against VLM scalar pseudo-labels:

$$\mathcal{L}_{abs}(\phi) = \mathbb{E}_{(x,g,\hat{r})} \left[(f_\phi^{abs}(\text{enc}(x), \text{enc}(g)) - \hat{r})^2 \right]$$

The RRM is trained via binary cross-entropy against VLM pairwise preference labels $y \in \{0, 1\}$:

$$\mathcal{L}_{rel}(\psi) = -\mathbb{E}_{(x_t, x_{t+1}, g, y)} [y \log \sigma(f_\psi^{rel}) + (1 - y) \log(1 - \sigma(f_\psi^{rel}))]$$

Both networks are 2-layer MLPs applied to frozen CLIP features. The mixing coefficient $\alpha = 0.4$ is selected by validation.

Learning or Inference Procedure

1. Collect a replay buffer of (x_t, x_{t+1}, g) triples from random-policy rollouts in the target environment.
2. Query VLM (LLaMA-Vision-3.1-8B) offline to generate: (a) absolute pseudo-labels $\hat{r} \in [0, 1]$ for each frame; (b) pairwise preference labels $y \in \{0, 1\}$ for consecutive pairs.
3. Train f_ϕ^{abs} and f_ψ^{rel} on the VLM labels.
4. Run PPO with combined reward $r_t = \alpha f_\phi^{abs} + (1 - \alpha) f_\psi^{rel}$, querying only the lightweight reward MLPs (no VLM calls during RL).
5. Optionally fine-tune reward models every 10^5 steps using additional VLM labels from the current policy’s rollouts.

What the Guarantee Says

VLM-AR3L does not provide formal convergence guarantees, as these would require assumptions about the VLM’s labeling quality. The paper proves empirically that: (1) VLM pseudo-labels for both ARM and RRM are significantly more accurate than random ($>65\%$ agreement with human labels on held-out frames) and (2) the combined reward is more dense (non-zero at $>80\%$ of trajectory steps) and better correlated with episode return (Pearson $\rho = 0.71$) than either signal alone ($\rho = 0.52$ for ARM, $\rho = 0.48$ for RRM).

Experimental Findings

Task	VLM-AR3L	EUREKA	VLM-RM	Random
Combat Spider	51.4%	37.2%	41.8%	8.3%
Milk Cow	62.3%	48.1%	50.6%	12.1%
Shear Sheep	57.8%	41.0%	44.3%	9.7%
Hunt Cow	48.9%	36.8%	39.2%	7.5%

Task success rates averaged over 100 evaluation episodes. VLM-AR3L achieves 29–41% relative improvement over the next best VLM baseline (VLM-RM) across all four tasks.

Ablations and Interpretation

ARM only: Removing the RRM and using only absolute rewards reduces success rates by 22–28% (average 24%). Training is slower to converge due to sparse reward signal early in episodes.

RRM only: Removing the ARM and using only relative rewards reduces success rates by 18–32% (average 25%). Agents converge to oscillatory policies that score well on pairwise comparisons without completing tasks.

Mixing coefficient α : Performance peaks at $\alpha = 0.4$ (40% absolute, 60% relative). Lower α increases reward density but loses task-completion grounding; higher α restores grounding but reintroduces sparsity.

VLM size: Replacing LLaMA-Vision-3.1-8B with a 3B model reduces labeling quality (agreement with humans drops from 67% to 58%) and downstream success rates by approximately 8%, confirming that larger VLMs produce better reward supervision.

Reference

Kuan-Chen Chen, Winston Chen, Wei-Fang Sun, Min-Chun Hu. **VLM-AR3L: Vision-Language Models for Absolute and Relative Rewards in Reinforcement Learning**. arXiv:2607.00483, July 2026. IJCAI 2026. <https://arxiv.org/abs/2607.00483>